

TUTORIAL 6

Write a C++ program to overload the /= operator for division of two complex numbers.

INPUT:

```
#include <iostream>
using namespace std;

class Complex {
    float real, imag;

public:
    Complex(float r = 0, float i = 0) {
        real = r;
        imag = i;
    }

    void operator/=(const Complex& c) {
        float denominator = c.real * c.real + c.imag * c.imag;

        float newReal = (real * c.real + imag * c.imag) / denominator;
        float newImag = (imag * c.real - real * c.imag) / denominator;

        real = newReal;
        imag = newImag;
    }

    void display() {
        cout << real << " + " << imag << "i" << endl;
    }
};

int main() {
    Complex c1(4, 6);
    Complex c2(2, 3);

    cout << "Before division: ";
    c1.display();

    c1 /= c2;

    cout << "After division: ";
    c1.display();

    return 0;
}
```

}

OUTPUT:

```
Before division: 4 + 6i  
After division: 2 + 0i
```